

EXPERIMENT – 2

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Section: 8

Title

File Copying Using Low-Level System Calls and System Call Tracing using strace

Aim

To implement a file-copy program using the low-level system calls open(), read(), write(), and close(), and trace the cat command using strace.

Theory

The open() system call is used to open a file, read() is used to read data from a file, write() is used to write data to a file, and close() is used to close an opened file. The strace command is used to trace the system calls made by a Linux command.

Algorithm

1. Start the program.
2. Open the source file using open().
3. Open or create the destination file using open().
4. Read data from the source file using read().
5. Write the data into the destination file using write().
6. Repeat the read and write operations until the entire file is copied.
7. Close both files using close().
8. Display the successful file-copy message.
9. Trace the cat command using strace.
10. Stop the program.

Code

```
k_anusri@DESKTOP-TM20BAQ:~$  
k_anusri@DESKTOP-TM20BAQ:~$ cat > exp2_filecopy.c <<'EOF'  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <stdlib.h>  
  
int main()  
{  
    int source, destination;  
    char buffer[1024];  
    ssize_t n;  
  
    source = open("input.txt", O_RDONLY);  
  
    if (source < 0)  
    {  
        perror("open");  
        return 1;  
    }  
  
    destination = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (destination < 0)  
    {  
        perror("open");  
        close(source);  
        return 1;  
    }  
  
    while ((n = read(source, buffer, sizeof(buffer))) > 0)  
    {  
        write(destination, buffer, n);  
    }  
  
    close(source);  
    close(destination);  
  
    printf("File copied successfully.\n");  
  
    return 0;  
}
```

Execution Screenshots

Screenshot 1 — Create input file

```
k_anusri@DESKTOP-TM20BAQ:~$ echo "This is Experiment 2 file copying using Linux system calls." > input.txt
```

Screenshot 2 — Compilation and execution

```
k_anusri@DESKTOP-TM20BAQ:~$ gcc exp2_filecopy.c -o exp2_filecopy && ./exp2_filecopy  
File copied successfully.
```

Screenshot 3 — Verify copied file

```
k_anusri@DESKTOP-TM20BAQ:~$ cat output.txt  
This is Experiment 2 file copying using Linux system calls.
```

Screenshot 4 — strace

```
k_anusri@DESKTOP-TM20BAQ:~$ strace cat input.txt
execve("/usr/bin/cat", ["cat", "input.txt"], 0x7ffc56544cf8 /* 25 vars */) = 0
brk(NULL) = 0x5f7a4faa2000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x70d489a70000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20763, ...}) = 0
mmap(NULL, 20763, PROT_READ, MAP_PRIVATE, 3, 0) = 0x70d489a6a000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d489a37000
mmap(0x70d489a3e000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x70d489a3e000
mmap(0x70d489a5f000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x70d489a5f000
mmap(0x70d489a66000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x70d489a66000
mmap(0x70d489a68000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x70d489a68000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d489a09000
mmap(0x70d489a0d000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x70d489a0d000
mmap(0x70d489a31000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x70d489a31000
mmap(0x70d489a35000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x70d489a35000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d4898e3000
mmap(0x70d4898f5000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x70d4898f5000
mmap(0x70d489995000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x70d489995000
mmap(0x70d489a07000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x70d489a07000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\250\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d489600000
mmap(0x70d489628000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x70d489628000
mmap(0x70d489628000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x70d489628000
mmap(0x70d4896c0000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x70d4896c0000
mmap(0x70d4896d4000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x70d4896d4000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcpre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699856, ...}) = 0
mmap(NULL, 761112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x70d489837000
mmap(0x70d48983a000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x70d48983a000
mmap(0x70d4898b6000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x70d4898b6000
mmap(0x70d4898e1000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x70d4898e1000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x70d489835000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x70d489832000
arch_prctl(ARCH_SET_FS, 0x70d489832b00) = 0
set_tid_address(0x70d489833128) = 572
set_robust_list(0x70d489832de0, 24) = 0
rseq(0x70d4898326a0, 0x20, 0, 0x53853955) = 0
mprotect(0x70d48980e000, 16384, PROT_READ) = 0
mprotect(0x70d4898e1000, 4096, PROT_READ) = 0
mprotect(0x70d489a07000, 4096, PROT_READ) = 0
mprotect(0x70d489a35000, 4096, PROT_READ) = 0
mprotect(0x70d489a66000, 4096, PROT_READ) = 0
mprotect(0x5f7a1f350000, 1462272, PROT_READ) = 0
mprotect(0x70d489ab5000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\xde\x6e\x5d\xaa\xfd\x7c\x73\xbe", 8, GRND_NONBLOCK) = 8
munmap(0x70d489a6a000, 20763) = 0
statfs("/sys/fs/selinux", {f_type=SVFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x85c68608, 0x525393c8]}, f_name_len=35, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NOEXEC|ST_NOATIME}) = 0
statfs("/selinux", 0x7fff4ad69980) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x5f7a4faa2000
brk(0x5f7a4fac3000) = 0x5f7a4fac3000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdf"..., 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
```

```
k_anusri@DESKTOP-TM20BA: ~$ strace -e open,read,write,close cat /usr/share/coreutils/locales/cat
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbdev\t...", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-... 1024) = 1024
read(3, "e 0 0\ndepts /dev/pts devpts rw,..., 1024) = 1024
read(3, " 0 0\nhugetlbfs /dev/hugepages hu..., 1024) = 1024
read(3, "s /mnt/wslg/run/user/1000 tmpfs "..., 1024) = 118
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x70d489645cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "5f7a1e9e7000-5f7a1ead9000 r--p 0"... 1024) = 1024
read(3, "0 08:30 13647 "... 1024) = 1024
read(3, "00 08:30 13650 "... 1024) = 1024
read(3, " /usr/lib/x86_64-linux"... 1024) = 1024
read(3, "0 13644 /us"... 1024) = 566
close(3) = 0
sched_getaffinity(572, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 0
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x70d489a6d000
mprotect(0x70d489a6d000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x70d489a6d000, ss_flags=0, ss_size=8192}, NULL) = 0
gettid() = 572
rt_sigaction(SIGBUS, {sa_handler=0x5f7a1ed0c0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x70d489645cb0}, NULL, 8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x5f7a1ed0c0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x70d489645cb0}, NULL, 8) = 0
prctl(PR_GET_AUXV, 0x7fff4ad69470, 0x200, 0, 0) = 448

k_anusri@DESKTOP-TM20BA: ~$ strace -e open,read,write,close cat /usr/share/coreutils/locales/cat
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
stat(3, {st_mode=S_IFREG|0644, st_size=4080, ...}) = 0
read(3, "# Common strings shared across a"... 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x70d489645cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x70d489645cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x70d489645cb0}, {sa_handler=0x5f7a1ed0c0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x70d489645cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x70d489645cb0}, {sa_handler=0x5f7a1ed0c0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x70d489645cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NLCR|TAB0|TAB8|BS0|VT0|FF0|OPOST|ONLCR, c_lflag=ISIG|ICANON|ECH0|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOW, ...}) = 0
brk(0x5f7a4fae4000) = 0x5f7a4fae4000
statx(AT_FDCWD, "input.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=60, ...}) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=60, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 60
splice(4, NULL, 1, NULL, 60, 0) = 60
This is Experiment 2 file copying using Linux system calls.
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(3) = 0
close(4) = 0
close(5) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x70d489a6d000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
```

Result

The file was copied successfully using the low-level system calls `open()`, `read()`, `write()`, and `close()`. The system calls made by the `cat` command were traced successfully using `strace`.